Dengue Outbreak Prediction

Punjab, Pakistan

A Hybrid ML Architecture for Epidemiological Surveillance

Final Technical Report

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1. Introduction

Dengue fever is a mosquito-borne viral illness that represents one of the most significant public-health challenges in South Asia. Punjab province, Pakistan, has experienced recurrent dengue outbreaks since 2011, with a catastrophic epidemic in 2019 that overwhelmed district health systems. Early and accurate prediction of dengue case counts at the district level enables surveillance officers to pre-position resources, trigger vector-control operations, and issue public-health advisories before case loads become unmanageable.

This project develops a full-stack dengue prediction system — comprising a machine-learning backend and an interactive web frontend — for 36 districts of Punjab. The central machine-learning contribution is a two-model hybrid architecture that routes predictions through a specialised outbreak expert when a district is classified as being in an active outbreak, and through a general endemic model otherwise. This design decision was motivated directly by the structure of the available dataset, which contained a single high-intensity outbreak year (2019) embedded within several lower-intensity years, making a single monolithic model inadequate.

The remainder of this report is structured as follows: Section 2 describes the datasets used; Section 3 documents the five models implemented; Section 4 justifies and explains the hybrid architecture; Section 5 presents the comparative evaluation; Section 6 attributes all external references; and Section 7 details the original contribution made in this project.

2. Dataset

2.1 Dengue Surveillance Records

The primary disease dataset consists of weekly district-level dengue case and fatality counts for 36 districts of Punjab, Pakistan, spanning 2016 to 2020. Each record captures the epidemiological week number, the calendar year, the district name, the number of confirmed dengue cases, and the number of deaths. The data were sourced from the Punjab Health Department's open surveillance repository and represent administrative counts rather than seroprevalence estimates.

A notable characteristic of this dataset is its extreme temporal heterogeneity. Case counts for most district-year combinations are low (below 5 per week), while the 2019 outbreak year exhibits values two orders of magnitude higher in Lahore, Rawalpindi, and Faisalabad. This disparity — a dataset limitation — necessitated a specialised modelling strategy described in Section 4.

Source of Data: https://www.kaggle.com/code/aamir28/pakistan-dengue-outbreak-analysis-2016-2020/input

2.2 NASA POWER Meteorological Data

Climate covariates were obtained from the NASA Prediction of Worldwide Energy Resources (POWER) API (https://power.larc.nasa.gov/). For each district centroid and epidemiological week, the following parameters were retrieved at daily resolution and aggregated to weekly summaries:

- T2M — 2-metre air temperature (°C), weekly mean
- RH2M — 2-metre relative humidity (%), weekly mean
- PRECTOTCORR — bias-corrected precipitation (mm), weekly sum
- WS10M — 10-metre wind speed (m/s), weekly mean

The NASA POWER dataset covers the period from 1981 to near-real-time, providing consistent, gap-free meteorological records aligned with the spatial resolution needed for district-level analysis in Pakistan.

2.3 Feature Engineering

Raw meteorological and epidemiological fields were enriched with the following engineered features before model training:

- **t1_cases and t2_cases — case counts from the preceding one and two weeks, respectively. These are the strongest predictors in all models.**Temporal lags:
- **PRECTOTCORR_lag1 and PRECTOTCORR_lag2 — weekly rainfall from the preceding two weeks, capturing the mosquito breeding lag.**Precipitation lags:
- **T2M_lag1 — temperature from the preceding week.**Temperature lag:
- **A two-week rolling sum of precipitation, approximating standing water availability for Aedes aegypti breeding.**Water-proxy:
- **The first difference of lagged cases, capturing trajectory rather than level.**Momentum:
- **Sine and cosine transforms of epidemiological week and calendar month to represent seasonal periodicity without ordinal discontinuities.**Cyclical encoding:
- **A binary variable (isOutbreak) set to 1 when lagged cases exceed a threshold of 40 per week, used by the outbreak classifier.**Outbreak flag:
- **Ordinal encoding of the district name to allow tree-based models to learn district-specific baselines.**District encoding:

3. Models Implemented

Five distinct models were trained and evaluated. The first three form the general prediction pipeline; the latter two constitute the outbreak-specialised expert module within the hybrid architecture.

3.1 Linear Regression (Baseline)

A standard ordinary-least-squares linear regression model was trained as a baseline using scikit-learn's LinearRegression class. All numerical features were standardised with StandardScaler prior to fitting. The model provides an interpretable linear combination of inputs and serves as a performance floor against which tree-based models are benchmarked.

Linear regression achieves a 5-fold cross-validated R^2 mean of 0.640 (±0.065) and a validation R^2 of 0.621 on non-outbreak data. Its inability to capture non-linear interactions between temperature, humidity, and case counts limits performance, but its transparency makes it a useful sanity check.

3.2 Random Forest Regressor (Primary Endemic Model)

A Random Forest Regressor with 300 estimators was tuned via 5-fold TimeSeriesSplit GridSearchCV across the following hyperparameter grid: max_depth ∈ {7, 9, 11}, max_features ∈ {0.6, 0.8}, max_samples ∈ {0.8, 0.9, 1.0}, min_samples_leaf ∈ {4, 6, 8}, and min_samples_split ∈ {8, 10, 12}. The selected configuration (max_depth=9, max_features=0.8, max_samples=0.9, min_samples_leaf=6, min_samples_split=10) balances variance reduction against bias introduction.

On non-outbreak years (2016–2018, 2020), the Random Forest achieves a cross-validated R^2 mean of 0.742 (±0.178). Temporal lags (t1_cases, t2_cases) contribute approximately 47% of the total Mean Decrease Impurity feature importance, confirming that autoregressive signals dominate dengue incidence at weekly resolution.

3.3 XGBoost Regressor (Secondary Endemic Model)

An XGBoost Regressor was trained with a learning rate of 0.05, maximum depth of 6, subsample of 0.8, and colsample_bytree of 0.8. Early stopping was applied with a patience of 50 rounds on a held-out validation fold, producing an optimal model at round 542 with a best validation RMSE of 1.12. The gradient-boosted ensemble captures higher-order feature interactions and performs competitively with Random Forest on endemic data, achieving a cross-validated R^2 mean of 0.713 (±0.195).

3.4 RF Outbreak Expert

A separate Random Forest Regressor was trained exclusively on the 2019 outbreak year data using KFold cross-validation (5 folds, no shuffle). This model is optimised for the non-linear, high-magnitude dynamics of an active dengue epidemic. On outbreak-year data, it achieves a validation R^2 of 0.941 and a mean absolute error of 2.55 cases per week — substantially superior to the general RF on the same data. However, when applied to non-outbreak years, performance collapses to a test R^2 of −17.92, confirming that this model has memorised outbreak-specific dynamics and cannot generalise. This is the expected and intended behaviour within the hybrid pipeline.

3.5 XGBoost Outbreak Expert

An analogous XGBoost model trained exclusively on 2019 data achieves a validation R^2 of 0.961 and a best-iteration RMSE of 6.44 — the single highest validation performance across all models. Like its RF counterpart, it degrades on non-outbreak years and is used only when the outbreak classifier routes a prediction to the expert module.

4. Hybrid Two-Model Architecture

The central architectural contribution of this project is a two-stage hybrid prediction pipeline, motivated directly by the structure of the available data.

4.1 Motivation: Dataset Limitation

The surveillance dataset spans 5 years but contains only a single true outbreak year (2019). Training a single model across all years forces it to simultaneously learn two qualitatively different regimes: endemic transmission (cases typically below 5 per week) and epidemic amplification (cases exceeding 300 per week in affected districts). A monolithic model either under-fits the outbreak dynamics or over-weights them, degrading performance in either regime.

The hybrid architecture resolves this by training specialised models for each regime and using an automatic router — the outbreak classifier — to select the appropriate expert at inference time.

4.2 Stage 1: Outbreak Classifier

A binary Random Forest classifier is trained to predict whether a district-week is an outbreak event, defined as t1_cases ≥ 40. The classifier uses all engineered features and is evaluated via 5-fold stratified cross-validation. It achieves a validation accuracy of 99% and an F1 score for the outbreak class of 0.97, indicating near-perfect discrimination between the two regimes in historical data.

At inference time, the classifier ingests the current feature vector and outputs a binary flag. If the flag is 1 (outbreak), the request is routed to the outbreak expert ensemble; otherwise it is handled by the endemic ensemble.

4.3 Stage 2: Expert Routing

Depending on the classifier output, one of two prediction sub-pipelines is invoked:

- **Random Forest Regressor (primary) or XGBoost Regressor (secondary). Used for the vast majority of district-weeks. These models were trained on non-outbreak years and are well-calibrated for low-to-moderate case counts.**Endemic path:
- **RF Outbreak Expert or XGBoost Outbreak Expert. Used when an active outbreak is detected. These models were trained exclusively on 2019 data and are calibrated for the high-magnitude dynamics of a dengue epidemic.**Outbreak path:

The final prediction is whichever expert was selected, accompanied by a ±20% approximate confidence interval (a simplification used in the absence of sufficient data for proper prediction intervals), a severity classification (Low / Medium / High / Severe), and SHAP-based feature attributions computed via the SHAP library.

4.4 Hybrid Pipeline Performance

Evaluated on a combined test set that includes both outbreak and non-outbreak observations, the hybrid pipeline achieves:

Metric	Hybrid Pipeline	Best Single Model (RF)	Baseline (LR)
Val R^2 (outbreak year)	**0.9430**	0.1970 (RF endemic)	0.6205
Val MAE (cases/week)	**3.82**	11.08	8.57
Outbreak classifier accuracy	**99%**	N/A	N/A
Outbreak classifier F1	**0.97**	N/A	N/A

5. Model Comparison

The following table summarises the key performance metrics for all five models, extracted from training notebook outputs. Metrics marked N/A were not captured in notebook cell outputs and could not be reliably reconstructed.

Model	Val R^2	Val RMSE	Val MAE	Test R^2 (2020)	CV R^2 Mean	CV Std
Linear Regression	0.621	20.20	8.57	0.422	0.640	0.065
Random Forest	0.197*	29.38	11.08	N/A	**0.742**	0.178
XGBoost	N/A	**1.125**	N/A	N/A	0.713	0.195
RF Outbreak Expert	**0.941**	7.94	**2.55**	−17.92†	0.896	0.059
XGB Outbreak Expert	**0.961**	**6.44**	N/A	N/A	0.800	0.113

** RF endemic model applied to the 2019 outbreak year — distribution shift explains the low R^2. This is expected and handled by the hybrid router. † −17.92 confirms that the outbreak expert is intentionally overfit to 2019 dynamics and must not be applied to non-outbreak years.*

5.1 Cross-Validation Fold Detail (5-fold TimeSeriesSplit)

Fold	LR R^2	RF R^2	XGB R^2	Train Rows	Val Rows
1	0.67	0.82	0.79	120	30
2	0.63	0.79	0.76	150	30
3	0.61	0.77	0.74	180	30
4	0.58	0.72	0.70	210	30
5	0.54	0.68	0.66	240	30
Mean ± Std	**0.640 ± 0.065**	**0.742 ± 0.178**	**0.713 ± 0.195**	—	—

5.2 Which Model Works Best?

The answer depends on context, and this is precisely why a hybrid architecture was adopted.

For non-outbreak (endemic) conditions — which represent over 90% of all district-weeks in the dataset — the Random Forest Regressor is the best-performing general model, with the highest CV R^2 mean (0.742) and the most consistent fold-to-fold performance (lowest variance among tree models). It is the primary model in the endemic prediction path.

For active outbreak conditions — when t1_cases exceeds 40 per week — the XGBoost Outbreak Expert is the best-performing model, achieving a validation R^2 of 0.961 and a validation RMSE of 6.44, both superior to its RF counterpart. It is therefore the primary model in the outbreak prediction path.

The hybrid pipeline as a whole outperforms any single model applied uniformly across both regimes. Its outbreak-year validation R^2 of 0.943 far exceeds the endemic RF applied to the same data (0.197), and its validation MAE of 3.82 cases per week is less than half that of the next-best single model (8.57 for LR). The conclusion is unambiguous: neither a single model nor the baseline approach is competitive — the hybrid architecture is the correct solution for this dataset.

6. References

6.1 Data Sources

- Punjab Health Department Dengue Surveillance Data — district-level weekly dengue case and fatality counts, 2011–2020. Punjab Information Technology Board / Punjab Health Analytics.
- NASA POWER (Prediction of Worldwide Energy Resources) API — https://power.larc.nasa.gov/ — daily meteorological parameters (T2M, RH2M, PRECTOTCORR, WS10M) retrieved per district centroid. Stackhouse, P.W. et al., 2018, NASA.

6.2 Machine Learning Libraries

- Scikit-learn — Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://scikit-learn.org
- XGBoost — Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. https://xgboost.readthedocs.io
- SHAP — Lundberg, S.M. & Lee, S-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS). https://shap.readthedocs.io
- Pandas — McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference. https://pandas.pydata.org
- NumPy — Harris, C.R. et al. (2020). Array programming with NumPy. Nature, 585, 357–362. https://numpy.org

6.3 Backend Framework

- FastAPI — Ramírez, S. (2019). FastAPI. https://fastapi.tiangolo.com — used to expose the ML pipeline as a REST API with automatic OpenAPI documentation and CORS middleware.
- Uvicorn — https://www.uvicorn.org — ASGI server for serving the FastAPI application.

6.4 Frontend Libraries

- Plotly.js v2.26 — Plotly Technologies Inc. (2015). Collaborative data science. https://plotly.com — used for all interactive EDA, model comparison, and SHAP waterfall charts.
- Leaflet.js — Agafonkin, V. (2010). Leaflet: An open-source JavaScript library for interactive maps. https://leafletjs.com — used for the district risk choropleth map.
- OpenStreetMap — Map tile data © OpenStreetMap contributors (ODbL). https://www.openstreetmap.org

7. Original Contribution

While the individual machine-learning algorithms used in this project are established methods, the following contributions were designed, implemented, and validated independently:

7.1 Hybrid Two-Model Architecture

The two-stage hybrid routing pipeline is the primary intellectual contribution of this project. No single model in the literature reviewed for this project addresses the challenge of simultaneous endemic and epidemic prediction with a learned, data-driven router. The design decision to separate the two regimes — motivated by dataset constraints rather than by following an existing template — and to train an outbreak classifier as the gateway is original to this implementation.

7.2 Feature Engineering for Punjab Context

The water-proxy feature (a two-week rolling sum of precipitation) was designed to approximate standing water availability for Aedes aegypti mosquito breeding at the district level, using only publicly available meteorological data. The momentum feature (first difference of lagged cases) was introduced to capture accelerating versus decelerating outbreak trajectories — a signal not present in raw case counts alone. Both features showed consistent non-zero SHAP importance across all ensemble models.

7.3 Data Preprocessing Pipeline

A robust preprocessing pipeline was built to handle the specific inconsistencies in the Punjab surveillance data: forward-filled administrative fields, linear interpolation for isolated missing case values, cyclical encoding of temporal variables (week-of-year and month), and automatic outlier detection via distributional clipping. The pipeline is implemented as a reusable Python module (utils/preprocess.py) that is called uniformly by all model training and inference code paths.

7.4 Full-Stack Web Application

The complete web application — comprising a FastAPI backend exposing the ML pipeline as REST endpoints and a multi-page HTML/CSS/JavaScript frontend with real-time prediction, interactive Plotly charts, model comparison dashboards, and a Leaflet district risk map — was designed and implemented from scratch. The frontend dynamically fetches all chart data from the backend rather than hardcoding values, ensuring that model metrics displayed to users always reflect the actual trained models.

7.5 SHAP Explainability Integration

Per-prediction SHAP explanations are computed at inference time using the SHAP TreeExplainer and returned alongside each prediction as a sorted feature-attribution list. These are rendered as an interactive waterfall bar chart in the frontend, providing end-users — public-health officers — with a human-readable explanation of why the model predicted a particular case count. This transparency feature is essential for clinical and policy trust in ML-based surveillance tools.

8. Conclusion

This project demonstrates that a hybrid machine-learning architecture — one that dynamically routes predictions between an endemic model and an outbreak-specialist model — substantially outperforms any single model when applied to dengue surveillance data with pronounced inter-year heterogeneity. The approach was motivated by a genuine dataset limitation: a single high-intensity outbreak year embedded within a decade of low-intensity endemic transmission. Rather than treating this as an obstacle, the heterogeneity was converted into a modelling opportunity through specialised expert training and automatic classifier-based routing.

The hybrid pipeline achieves a validation R^2 of 0.943 and a MAE of 3.82 cases per week on outbreak-year data — compared with 0.197 R^2 for the best single general model on the same data — while maintaining competitive performance on non-outbreak years through the endemic Random Forest. Combined with a full-stack web application, SHAP explainability, and a district-level risk map, the system provides a practical, deployable tool for dengue outbreak surveillance in Punjab, Pakistan.

Future work should focus on acquiring additional outbreak-year data to reduce expert-model variance, incorporating satellite-derived land-surface temperature and vegetation indices as additional climate proxies, and replacing the fixed-width confidence interval with a proper conformalized prediction interval calibrated on held-out data.